

# Notes on Sparse Multivariate Methods

Multivariate Data Analysis (MAE0330)

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# Summary

1. The Big- $p$  Problem ( $n \ll p$ )
2. Sparse Principal Component Analysis
3. Sparse Discriminant Analysis
4. Sparse Canonical Correlation Analysis
5. References

# The Big- $p$ Problem

# The Big- $p$ Problem ( $n \ll p$ ) I

When we have a data set with a very large number of variables (parameters)  $p$  in relation to the number of observations (individuals)  $n$ , that is,  $n \ll p$ , we commonly say that we have a **big- $p$  problem** (sometimes **big- $p$ , little- $n$** ).

The techniques of Principal Component Analysis (PCA), Discriminant Analysis (DA) and Canonical Correlation Analysis (CCA) work well in the task of dimensionality reduction for the classical case ( $n > p$ ), however, in the case where  $n \ll p$ , these techniques are not convenient.

An alternative to overcome this problem is the use of **sparse methods**, which are adaptations of these techniques for the case  $n \ll p$  using **penalties** and **regularizations**.

# The Big- $p$ Problem ( $n \ll p$ ) II

Using penalization and regularization techniques we obtain the sparse versions of PCA, DA and CCA (which we will discuss in the next slides):

- ▶ Sparse Principal Component Analysis (*Sparse PCA* or sPCA);
- ▶ Sparse Discriminant Analysis (*Sparse DA* or sDA);
- ▶ Sparse Canonical Correlation Analysis (*Sparse CCA* or sCCA).

**Sparsity:** A vector  $x$  (or matrix  $X$ ) is said to be sparse if many of its entries  $x_i$  ( $x_{ij}$ ) are equal to zero.

# Sparse Principal Component Analysis

# Principal Component Analysis

## A Review of Principal Components and Principal Coordinates

Let  $X = X_{n \times p}$  be a data matrix. We have already seen that we can obtain the  $k$ -th *principal component* (PC), denoted by  $Z_k$ , by the spectral decomposition of the covariance matrix, i.e.  $\Sigma = VDV' \Rightarrow Z_k = XV_k$ , where  $V_k$  are the column vectors of  $V$  (eigenvectors).

Equivalently,  $Z_k$  can be obtained through the singular value decomposition (SVD) of  $X$ , i.e.  $X = U\Lambda^{1/2}V'$ :

$$Z_k = U_k \Lambda_{kk}^{1/2}, \quad (1)$$

where  $U_k$  are the column vectors of  $U$  and  $\Lambda_{kk}^{1/2}$  are the singular values. In this case,  $Z_k$  are called *principal coordinates* (PCo). We can obtain equation (1) using *multidimensional scaling* from the matrix of *Euclidean* distances between the observations.

**Equivalence between PC and PCo:** The PCo analysis of the Euclidean distance matrix ( $n \times n$  matrix) is equivalent to the PC analysis of the covariance matrix ( $p \times p$  matrix).

# Sparse Principal Component Analysis

When we have a problem  $n \ll p$ , the disadvantage of performing “classical” PCA comes from the fact that the PCs are linear combinations of all  $p$  input variables, and since the number  $p$  is very large, the computational effort required to perform the computations is exaggeratedly large. An alternative to this problem is to use **Sparse Principal Component Analysis**.

For  $n \ll p$ , the interest is to make a **selection of the most important variables**, for the purpose of reducing the dimensionality (reduce  $p$ ). Therefore, more than obtaining the reduction vectors through PCs, we want to obtain this reduction by means that **penalize** those variables that must be eliminated (brought to null), that is, obtain eigenvectors  $V$  that assign zero load to some variables. To do this, we use regression algorithms: **penalized solutions** and **regularized solutions**.



# Sparse Principal Component Analysis - LASSO I

## Penalized PC (LASSO)

We want to predict the principal components ( $Z_k$ ) based on linear combinations of the data matrix  $X$  with vectors  $\beta$  (i.e., we want to find  $\beta$ 's such that  $X\beta \approx Z_k$ ).

*Lagrangian Penalty:*

$$\hat{\beta}_{lasso} = \arg \min_{\beta} \{ \|Z_k - X\beta\|_2^2 + \lambda \|\beta\|_1 \}, \quad (2)$$

where  $\|\cdot\|_2$  is the Euclidean norm,  $\|\cdot\|_1$  is the  $\ell^1$  norm and  $\lambda$  is the penalty parameter: if  $\lambda \rightarrow 0 \Rightarrow$  Least Squares solution, if  $\lambda \rightarrow \infty \Rightarrow \beta \rightarrow 0$ . The PCs  $Z_k$  of equation (2) are known, obtained by multidimensional scaling in  $\mathbb{R}^{n \times n}$  (i.e.  $Z_k$  is the  $k$ -th principal coordinate).

**Limitation:** The number of non-zero  $\beta$  is at most  $n$ .

# Sparse Principal Component Analysis - LASSO II

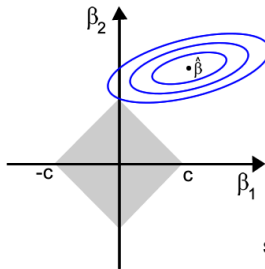
## Penalty in the restriction form:

Similarly, we can formalize the model by explaining the restriction in the vector  $\beta$ . For the two-dimensional case, we have:

$$\hat{\beta}_{2 \times 1} = \arg \min_{\beta} \sum_{i=1}^n (Z_{ik} - X_i' \beta)^2,$$

$$|\beta_1| + |\beta_2| \leq c.$$

Pictorially:



Penalized PC (lasso)

# Sparse Principal Component Analysis - Ridge Regression I

## Regularized PC (Ridge Regression)

Replacing the  $\ell^1$  norm with the Euclidean norm in the LASSO model, we obtain a *regularized* estimate for  $\beta$  known as **Ridge Regression**:

*Lagrangian regularization:*

$$\hat{\beta}_{ridge} = \arg \min_{\beta} \{ \|Z_k - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \}, \quad (3)$$

where  $\lambda$  is the regularization parameter: if  $\lambda \rightarrow 0 \Rightarrow$  Least Squares solution, if  $\lambda \rightarrow \infty \Rightarrow \beta \rightarrow 0$ . The PCs  $Z_k$  of equation (3) are known, obtained by multidimensional scaling (in  $\mathbb{R}^{n \times n}$ ).

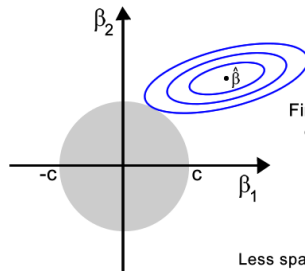
# Sparse Principal Component Analysis - Ridge Regression II

## Regularization in the restriction form:

Analogously, we can formalize the model by explaining the restriction in the vector  $\beta$ .  
For the two-dimensional case, we have:

$$\hat{\beta}_{2 \times 1} = \arg \min_{\beta} \sum_{i=1}^n (Z_{ik} - X_i' \beta)^2,$$
$$\beta_1^2 + \beta_2^2 \leq c.$$

Pictorially:



Regularized PC (Ridge Regression)

# Sparse Principal Component Analysis - Elastic Net

## Penalized and Regularized PC (Elastic Net; Zou et al. [8])

The following model, known as [Elastic Net](#), is a generalization of the LASSO model, and was introduced by Zou and Hastie [8]:

$$\hat{\beta}_{en} = \arg \min_{\beta} \{ ||Z_k - X\beta||_2^2 + \lambda_1 ||\beta||_2^2 + \lambda_2 ||\beta||_1 \}, \quad (4)$$

where  $\lambda_1$  is the regularization parameter and  $\lambda_2$  is the penalty parameter. We can fix  $\lambda_1$  and  $\lambda_2$  or obtain them by cross-validation.

**Advantage:** All variables can be selected (there is no limitation on the number of non-zero charges).

# Sparse Principal Component Analysis

From the estimates of the  $\beta$  vectors obtained by one of the previous models  $(\hat{\beta}_{lasso}, \hat{\beta}_{ridge}, \hat{\beta}_{en})$ , we obtain the approximation for the principal components  $Z_k$ :

$$\hat{Z}_k = X \hat{V}_k, \quad (5)$$

where

$$\hat{V}_k = \frac{\hat{\beta}}{\|\hat{\beta}\|_2}. \quad (6)$$

For more details on sparse PCA, see Zou et al. [9] and Hastie et al. [4].

# Sparse Principal Component Analysis - Example (Breast.TCGA)

## Implementation in R - ElasticNet

Sparse PCA is implemented in R in the [elasticnet](#) [10] package. In the following example, we use data from R's Bioconductor (Breast.TCGA).

**Data:** Breast.TCGA: Three databases (X1=mRNA, X2=miRNA and X3=Protein) evaluated on 150 individuals:

**X1=mRNA:** (breast.TCGA\$data.train\$mRNA)  $n = 150$ ,  $p = 200$ . ( $n \ll p$ )

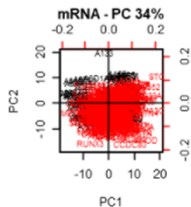
**X2=miRNA:** (breast.TCGA\$data.train\$mRNA)  $n = 150$ ,  $p = 184$ . ( $n \ll p$ )

**X3=Protein:** (breast.TCGA\$data.train\$mRNA)  $n = 150$ ,  $p = 142$ . ( $n > p$ )

**Subtypes of cancers:** (breast.TCGA\$data.train\$subtype) basal: 45; Her2: 30; LumA: 75.

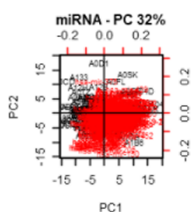
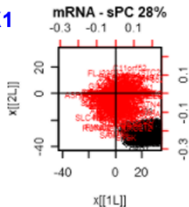
# Sparse Principal Component Analysis - Biplots

Principal Coordinates (PCo)

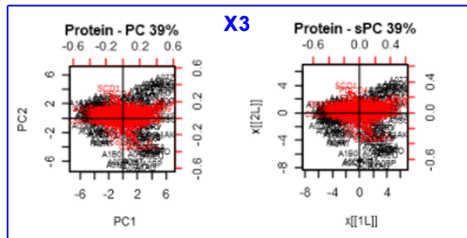
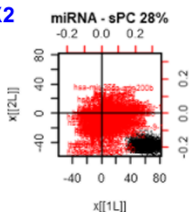


X1

Sparse PC



X2



X3

The biplots for the X3 (Protein) data are identical, because in this case we have  $n > p$ , that is, it is not a big-p problem.

## PCo: prcomp() (Stats)

```
x1.pc <- prcomp(x1)
biplot(x1.pc$x[,1:2],
       x1.pc$rotation[,1:2],
       var.axes=TRUE, main="mRNA -
       PC 34%")
```

## Sparse PCs: SPCA() (ElasticNet)

```
x1.spca <- spca(x1, K = 2,
                 type = "predictor",
                 sparse = "penalty",
                 para = rep(1e-05, 2))
```



# Sparse Discriminant Analysis

# Discriminant Analysis

In discriminant analysis (Fisher linear), as we have grouped observations, we consider the decomposition of the covariance matrix into two components: covariance due to the between groups effect and covariance due to the within groups effect,  $\Sigma_{p \times p} = \Sigma_{B_{p \times p}} + \Sigma_{W_{p \times p}}$ . From this, we are interested in solving the following optimization problem:

$$\max_l \frac{l' \Sigma_B l}{l' \Sigma_W l}. \quad (7)$$

In other words, we want to find vectors  $l$  such that maximize the ratio (7). This problem is equivalent to finding the eigenvalues and eigenvectors of  $\Sigma_W^{-1} \Sigma_B$ , which is equivalent to finding solutions of the determinant equation:

$$|\Sigma_W^{-1} \Sigma_B - \lambda I_p| = 0. \quad (8)$$

We assume homoscedasticity in the groups.

# Sparse Discriminant Analysis I

However, in the case where  $n \ll p$  (big- $p$ ), the inverse of the covariance matrix within groups,  $\Sigma_W$ , does not exist (it is singular), since the rank of this matrix is in maximum  $n$ . An alternative to correct the problem of the incomplete rank of  $\Sigma_W$  is to use *Sparse Discriminant Analysis* (sDA). In the following, we present the sDA models proposed by Witten et al. [7] and Clemmensen et al. [1].

## Regularization through $\Omega$ matrix

We can find a positive-definite diagonal matrix  $\Omega$  such that

$$|(\Sigma_W + \Omega) - dI_p| = 0; \quad d > 0. \quad (9)$$

If all the eigenvalues of a matrix are positive, then it is invertible (non-singular). Algorithms for obtaining the matrix  $\Omega$  are discussed in Hastie et al. [3].

## Sparse Discriminant Analysis II

Hence our optimization problem,  $\max_{\beta_k} \frac{\beta'_k \Sigma_B \beta_k}{\beta'_k \Sigma_W \beta_k}$ , becomes:

$$\max_{\beta_k} \frac{\beta'_k \Sigma_B \beta_k}{\beta'_k (\Sigma_W + \Omega) \beta_k}. \quad (10)$$

Equivalently, we can find a positive-definite matrix  $\Omega$  such that the discriminant vectors of the optimization problem

$$\max_{\beta_k} \{\beta'_k \Sigma_B \beta_k\}, \quad (11)$$

where  $\beta'_k (\Sigma_W + \Omega) \beta_k = 1$  and  $\beta'_k (\Sigma_W + \Omega) \beta_l = 0$ ,  $\forall l < k$ , can be calculated, even when  $n \ll p$ .

## Sparse Discriminant Analysis III

Furthermore, we want the load vectors (discriminant vectors)  $\beta_k$  to be *sparse*. A way to obtain these vectors is by applying the  $\ell^1$  (LASSO) penalty to the previous optimization problem, resulting in the following problem:

$$\max_{\beta_k} \{\beta_k' \Sigma_B \beta_k - \gamma \|\beta_k\|_1\}, \quad (12)$$

where  $\beta_k'(\Sigma_W + \Omega)\beta_k = 1$  and  $\beta_k'(\Sigma_W + \Omega)\beta_l = 0, \forall l < k$ , can be calculated, even when  $n \ll p$ . This method was proposed by Witten and Tibshirani [7].

## Sparse Discriminant Analysis IV

Another sparse discriminant analysis (sDA) method, proposed by Clemmensen et al. [1], is defined sequentially as follows. The  $k$ -th pair  $(\theta_k, \beta_k)$  is the solution to the problem:

$$\min_{\beta_k, \theta_k} \left\{ \|G\theta_k - X\beta_k\|_2^2 + \gamma \beta_k' \Omega \beta_k + \lambda \|\beta_k\|_1 \right\}, \quad (13)$$

where  $\frac{1}{n} \theta_k' G' G \theta_k = 1$  and  $\theta_k' G' G \theta_l = 0, \forall l < k$ , where  $\theta_{k_{N \times 1}}$  are the group weight vectors,  $G_{n \times N}$  is a group incidence matrix (composed by 0's and 1's) and  $\gamma$  and  $\lambda$  are the non-negative regularization and penalization parameters. The  $\ell^1$  penalty on  $\beta_k$  results in sparsity when  $\lambda$  is large.

The  $\beta_k$  vector that resolves (13) is called the  $k$ -th discriminant vector of the sDA.

# Sparse Discriminant Analysis V

To solve (13), we use a simple iterative algorithm to obtain a local optimum for (13). The algorithm involves keeping  $\theta_k$  fixed and optimizing with respect to  $\beta_k$ , and keeping  $\beta_k$  fixed and optimizing with respect to  $\theta_k$ . For fixed  $\theta_k$ , we obtain:

$$\min_{\beta_k} \left\{ \|G\theta_k - X\beta_k\|_2^2 + \gamma\beta_k'\Omega\beta_k + \lambda\|\beta_k\|_1 \right\}. \quad (14)$$

Note that for  $\Omega = I$ , (14) is exactly an ElasticNet problem.

# Sparse Discriminant Analysis - Example (Breast.TCGA)

## Implementation in R - sparseLDA

sDA is implemented in R in the [sparseLDA](#) package [12]. In the following example, we use the same data (`Breast.TCGA`) that was used in the previous example.

Remember that the data from this set is classified into [three groups](#) of subtype of cancers:

**G1: basal:** 45; **G2: Her2:** 30; **G3: LumA:** 75.

In other words, we have  $N = 3$  groups,  $G = G1 \cup G2 \cup G3$ , with a total of  $\#(G) = 45 + 30 + 75 = 150$  individuals.



# Sparse Discriminant Analysis - Scores e Loads

## X1 (mRNA):

Scores (discriminant functions):

|      | $x\hat{\beta}_1$ | $x\hat{\beta}_2$ |
|------|------------------|------------------|
|      | LD1              | LD2              |
| A0FJ | 1.822563         | -0.90566072      |
| A0G0 | 1.817337         | -0.32230828      |
| A0DA | 2.772156         | -2.23228938      |
| A0B3 | 2.229491         | -0.74279549      |
| A0I2 | 3.422815         | -2.11782785      |
| A0RT | 2.787478         | -1.96265989      |
| A131 | 1.487769         | 2.00316138       |
| A124 | 1.494891         | -0.88192122      |
| A1B6 | 2.224953         | 0.05620507       |
| A1AZ | 3.487032         | 0.09911661       |
| A0YM | 3.206956         | -1.17263109      |
| A04P | 1.871571         | -0.46004985      |
| A04T | 3.113374         | -0.41541073      |
| A0AT | 2.106453         | 0.81464297       |
| ...  |                  |                  |

Variable loads:

| $\hat{\beta}_1$ | $\hat{\beta}_2$ |
|-----------------|-----------------|
| 0.00000000      | -0.6549641      |
| -1.14725442     | 0.00000000      |
| -1.48943562     | 0.00000000      |
| 0.06294696      | 0.00000000      |
| 0.00000000      | -0.7050653      |
| 0.00000000      | 2.0153849       |
| ...             |                 |

**Discriminant variables:  
sda() (sparseLDA)**

```
sda.x1 <- sda(x1t, yt,  
lambda = 1e-6, stop = -3,  
maxIte = 25, trace = TRUE)
```

Group weight matrix:

| $\hat{\theta}_1$ | $\hat{\theta}_2$ |
|------------------|------------------|
| 1.3460511        | -0.7163423       |
| 0.3229238        | 2.0027743        |
| -0.9289263       | -0.3495289       |

Incidence matrix

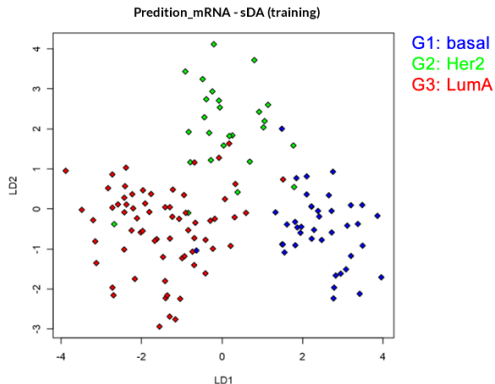
$G_{n \times N} = G_{150 \times 3}$

|     |   |   |
|-----|---|---|
| 1   | 0 | 0 |
| 1   | 0 | 0 |
| 1   | 0 | 0 |
| 1   | 0 | 0 |
| 1   | 0 | 0 |
| 1   | 0 | 0 |
| ... |   |   |

# Sparse Discriminant Analysis - Classification

## X1 (mRNA):

Representation of predicted groups for training data in sparse discriminant variables (LD1, LD2):



Classification accuracy (training data):

```
class.vector: Basal = 1; Her2 = 2, LumA = 3
```

```
class.vector Basal Her2 LumA
1 38 1 1
2 1 21 4
3 1 4 62
```

```
yt
Basal Her2 LumA
40 26 67
```

Accuracy:

0.909774436090226

# Sparse Canonical Correlation Analysis

# Classical Canonical Correlation Analysis

Consider the data matrix  $X_{n \times (p+q)} = (X_{1_{n \times p}} \ X_{2_{n \times q}})$ . Let  $X_{p \times 1}^1$  and  $X_{q \times 1}^2$  be the original variables such that:

$$\begin{pmatrix} X^1 \\ X^2 \end{pmatrix} \sim iid \left( \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right).$$

Canonical correlation analysis aims to solve the following optimization problem: find vectors  $a$ ,  $b$  such that maximize the correlation coefficient  $Corr(a'X^1, b'X^2)$ , that is,

$$\max_{a,b} \left\{ \frac{Cov(a'X^1, b'X^2)}{\sqrt{Cov(a'X^1)}\sqrt{Cov(b'X^2)}} \right\} = \max_{a,b} \left\{ \frac{a'\Sigma_{12}b}{\sqrt{a'\Sigma_{11}a}\sqrt{b'\Sigma_{22}b}} \right\}. \quad (15)$$

# Sparse Canonical Correlation Analysis

However, when we have  $n \ll p$  and  $n \ll q$ , occurs the impasse that the matrices  $\Sigma_{11}$  and  $\Sigma_{22}$  are singular (non-invertible). Furthermore, classical CCA results in vectors  $U, V$  that are not sparse, and these vectors are not unique if  $p$  or  $q$  exceeds  $n$ . An alternative to overcoming this problem is to use *Sparse Canonical Correlation Analysis* (sCCA).

For sCCA, Witten et al. [6] proposed a penalized solution for the singular value decomposition (SVD) of matrices, called **Penalized Matrix Decomposition (PMD)**.

This method does not involve the inverses of the covariance matrices, but the cross-product matrix  $X_1'X_2$ . Applying PMD to this cross-product matrix, we obtain a **penalized** method for CCA.

To this aim, we will work with **centered** and scaled columns  $X_1$  and  $X_2$ . Also, we will use **sample correlation**, which, for centered  $x, y \in \mathbb{R}^m$ , is given by:

$$\text{cor}(x, y) = \frac{x'y}{\sqrt{x'x}\sqrt{y'y}}. \quad (16)$$

# Sparse Canonical Correlation Analysis - PMD I

## Penalized Matrix Decomposition (PMD)

Consider the SVD decomposition,  $X = UDV'$ ,  $U'U = I_n$ ,  $V'V = I_p$ . Let  $U_k$  and  $V_k$  be the column vectors of  $U$  and  $V$ , respectively, and  $d_k$  be the diagonal elements of  $D$ . In [6], the following generalization of the approximation of  $X$  through least squares (first proposed by Eckart et al. [2]) was proposed:

$$\min_{U_k, V_k, d_k} \{ \|X - d_k U_k V_k'\|_2^2 \}, \quad (17)$$

with restrictions  $\|U_k\|_2^2 \leq 1$ ,  $\|U_k\|_1 \leq c_1$ ;  $\|V_k\|_2^2 \leq 1$ ,  $\|V_k\|_1 \leq c_2$ .

# Sparse Canonical Correlation Analysis - PMD II

In [6], as a corollary of theorem 2.1, it was verified that the previous problem is equivalent to the following maximization problem:

$$\max_{U_k, V_k} \{U_k' X V_k\}, \quad (18)$$

with restrictions  $\|U_k\|_2^2 \leq 1$ ,  $\|U_k\|_1 \leq c_1$ ;  $\|V_k\|_2^2 \leq 1$ ,  $\|V_k\|_1 \leq c_2$ .

One solution is to fix  $U$  and get  $V$ ; fix  $V$  and get  $U$ :

- Fixed  $V_k$ :  $\max_{U_k} \{U_k' X V_k\}$ ;  $\|U_k\|_2^2 \leq 1$ ,  $\|U_k\|_1 \leq c_1$ ,  $1 \leq c_1 \leq \sqrt{n}$ ;
- Fixed  $U_k$ :  $\max_{V_k} \{U_k' X V_k\}$ ;  $\|V_k\|_2^2 \leq 1$ ,  $\|V_k\|_1 \leq c_2$ ,  $1 \leq c_2 \leq \sqrt{p}$ .

This algorithm is spelled PMD( $L_1$ ,  $L_1$ ).

# Sparse Canonical Correlation Analysis - Penalized sCCA via PMD

Sparse canonical correlation analysis uses the  $\text{PMD}(L_1, L_1)$  algorithm ([sCCA Penalized via PMD](#)), considering the SVD decomposition of the matrix  $X_1'X_2$  (sample covariance matrix), as follows (for the norm  $\ell^1$ ):

$$\max_{a_k, b_k} \{(X_1 a_k)' X_2 b_k\} = \max_{a_k, b_k} \{a_k' X_1' X_2 b_k\}, \quad (19)$$

with restrictions  $a_k' X_1' X_1 a_k \leq 1$ ,  $\|a_k\|_1 \leq c_1$  e  $b_k' X_2' X_2 b_k \leq 1$ ,  $\|b_k\|_1 \leq c_2$ .

Assuming that for high-dimensional data the diagonal covariance matrix can be adopted (CCA-P Diagonal), the previous restrictions become:

$a_k' X_1' X_1 a_k = a_k' a_k \leq 1$ , pois  $X_1' X_1 = I_p$ , e  $b_k' X_2' X_2 b_k = b_k' b_k \leq 1$ , pois  $X_2' X_2 = I_q$ .

Another approach to sCCA can be found in Suo et al. [5].



# Sparse Canonical Correlation Analysis - Example (Breast.TCGA)

## Implementation in R - PMA

sCCA is implemented in R in the [PMA](#) (*Penalized Multivariate Analysis*) package [11]. In the following example, we use the same data (`Breast.TCGA`) that was used in the previous two examples. However, we now want to analyze the pairwise correlation of the three multivariate databases:

**Integration X1\_X2:**  $\max_{a,b} \{ \text{cor}(X_1 a, X_2 b) \}$

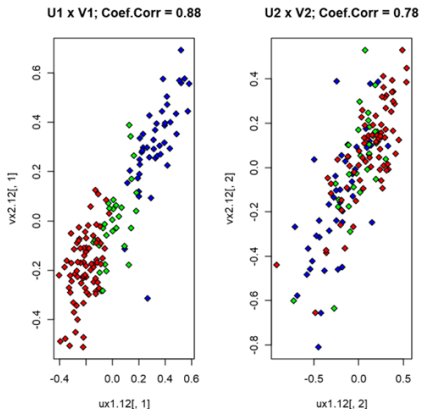
**Integration X1\_X3:**  $\max_{a,b} \{ \text{cor}(X_1 a, X_3 b) \}$

**Integration X2\_X3:**  $\max_{a,b} \{ \text{cor}(X_2 a, X_3 b) \}$

# Sparse Canonical Correlation Analysis - sCCA on X1\_X2

## Integration X1\_X2

Observations represented on canonical axes U1 x V1 e U2 x V2:



Sparse canonical vectors:

u and v  
maximize  $u'X_1'X_2v$

$v = (v_1, v_2):$

```
0.0000000 0
0.0000000 0
0.0000000 0
0.0000000 0
0.0000000 0
0.1562996 0
...
```

$u = (u_1, u_2):$

```
0 0
0 0
0 0
0 0
0 0
0 0
...
```

**sCCA via PMD:  
CCA() (PMA)**

```
scca.12 <- CCA(x1,x2,typeex=
"standard",typez="standard",
K=2)
```

Canonical variables:  $U = (U1 \ U2) = (X1*u1 \ X1*u2)$

$V = (V1 \ V2) = (X2*v1 \ X2*v2)$

Canonical correlation coefficients:

```
Cor(X1*u1, X2*v1), Cor(X1*u2, X2*v2):
0.88443973794229 0.779709063287576
```

# References I

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